

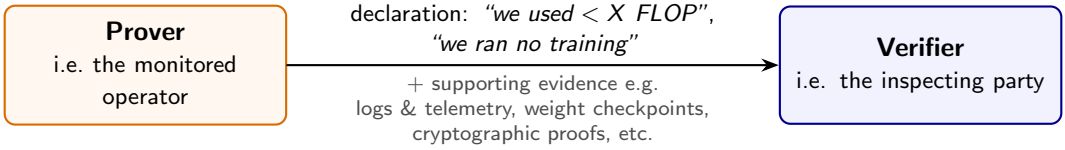
Can Power Draw Constrain Covert Compute?

Tom Kimpson
University of Melbourne & MATS
with Mauricio Baker, Emlyn Graham, Anjay Friedman, Maximus Rafla, Coby Joseph

MATS, August 2026

Why Verify Compute?

- AI governance agreements require **verification**.
- An outside party ("verifier") must be able to check what a compute operator ("prover") *actually ran*.

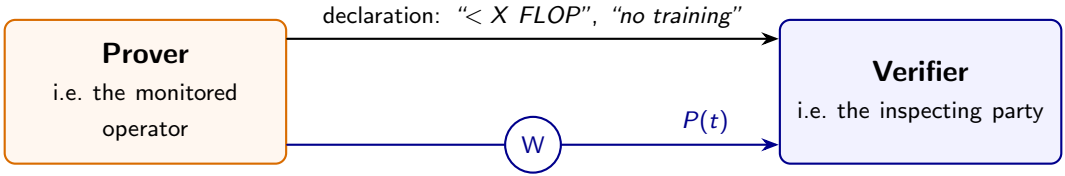


Verification Must Survive an Adversary

- The prover may **not behave honestly** — try to hide compute, try to run training
- "Evidence" is provided by the (cheating?) prover based on *silicon the prover controls*, e.g. utilisation counters, on-chip clocks, attested firmware, etc.
- **Off-chip analogue sensors** (power, EM, acoustic, thermal) bypass that: physically separate, retrofittable, securable against tampering.

This Work: Power Channel

We focus on one analogue channel: **power draw** — *every operation costs energy*.



What kind of compute ran?
time-resolved profile:
training cadence, periodicity

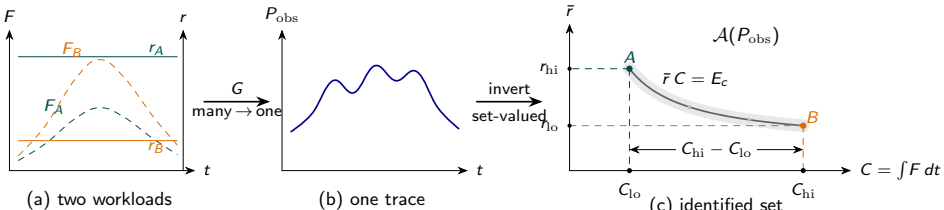
companion work — not this talk

How much compute ran?
integrated energy:
can a trace bound total FLOP?

this talk

What The Meter Sees

$$\underbrace{P(t)}_{\text{what the meter sees}} = \underbrace{r(t)}_{\text{pJ/op: prover's choice, invisible}} \cdot \underbrace{F(t)}_{\text{op/s: what the verifier wants}} + \underbrace{P_0(t)}_{\text{overhead: idle, cooling}}$$

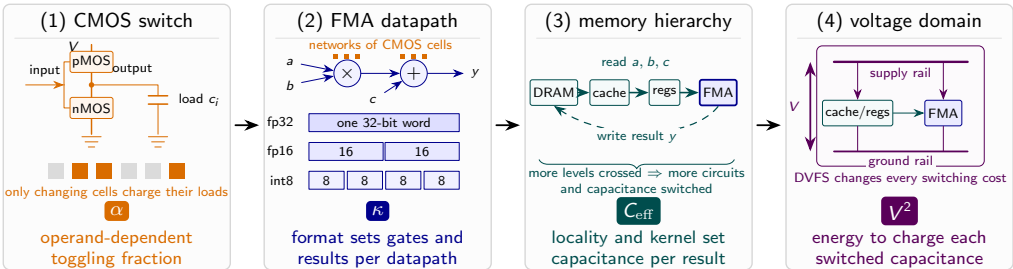


(a) two workloads (b) one trace (c) identified set

Goal: bound the width $C_{\text{hi}} - C_{\text{lo}} = C_{\text{hi}} - C_{\text{dec}} = C_{\text{covert}}^{\max}$

What Sets the Exchange Rate r?

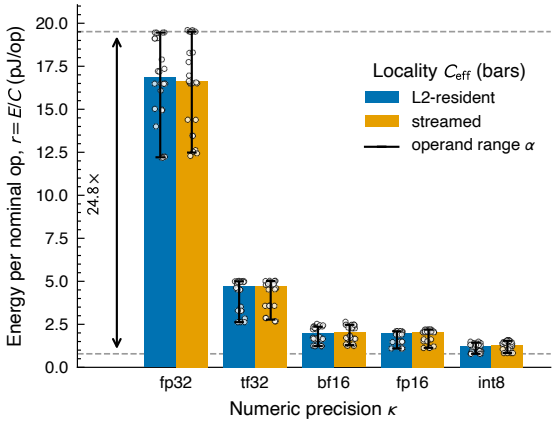
The cost of an operation is **not a device constant** — it is a property of the *execution*:



$$r = \alpha \cdot \kappa \cdot C_{\text{eff}} \cdot V^2$$

How Wide Are the Bands? Measured on an A100

We measured the bands directly on an NVIDIA A100:

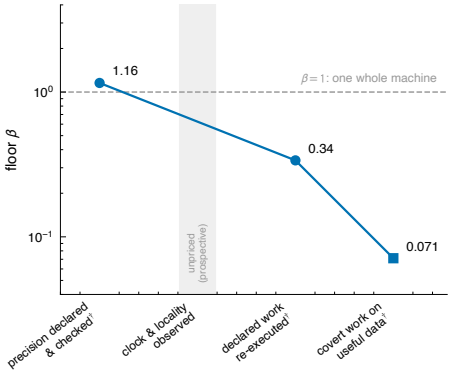


- **The declared band:** cost per op spans **24.8×** (0.79–19.5 pJ/op).
  per-factor widths: κ: 13.5×, α: 1.9×, C_{eff}: 1.04×, V^2: ≈1.5× (assumed)
- **The overhead band:** idle draw wanders over $P_0 \in [60, 101]$ W — a 41 W band the verifier must concede.

How Much Compute Can Hide?

Define $\beta \equiv C_{\text{covert}}^{\max} / C_{\max}$

- $\beta = 1.16$ — just power + a checked precision declaration
- $\beta = ?$ — clock & locality: not measurable with power alone, nor by our experiments
- $\beta = 0.34$ — re-execute the declared work (a hardware gap persists)
- $\beta = 0.07$ — restrict covert work to being *useful*



Limitations

- One vendor, one GPU model, one cluster: the numbers are A100-specific (but maths/physics is general).
- DVFS never swept (platform denied clock locking)
- Energy read from on-die NVML telemetry; in practice an off-chip meter adds its own uncertainties
- Single device: facility scale is a multi-resource allocation problem (shared cooling, work placement, etc.).
- Everything is time-integrated; richer trace models, multiple channels, or attested telemetry are outside it and may do better.

Summary

- **Verification** is important for AI governance.
- **Off-chip analogue measurements** (e.g. power) can be used for verification.
- Power measurements + prover declarations + reasonable assumptions constrain covert compute to $C_{\text{covert}} \lesssim 7\%$ of machine capacity.
- Extending to more realistic, representative **hardware** is the next goal.

Can Power Draw Constrain Covert Compute? — Kimpson, Baker, Graham, Friedman, Rafla, Joseph

tom.kimpson@unimelb.edu.au